

Pallor:
Diagnosis of anaemia
Nutritional anaemia
Anaemia of systemic diseases

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Contents

1. Clinical presentation and physical examination of an anaemic patient
2. Investigations and causes of anaemia
3. Nutritional anaemia
4. Anaemia of chronic illness

Reference: Essential Haematology Hoffbrand & Pettit

Which is the most important source of vitamin B12
in our body?

1. Animal and dairy products
2. Bacterial synthesis from intestines
3. Fermented food products
4. Green vegetables
5. Sunlight exposure

Which is the most common cause of vitamin B12 deficiency anaemia ?

1. A strict vegetarian diet
2. Inflammatory bowel disease
3. Pernicious anaemia
4. Prolonged use of metformin
5. Prolonged use of proton pump inhibitors

Which of the following tests is commonly used in the investigation of vitamin B12 deficiency anaemia?

1. RBC vitamin B12 level
2. Serum homocysteine level
3. Serum methylmalonic acid
4. Serum holotranscobalamin level
5. Serum total vitamin B12 level

Which of the following is the most common cause of iron deficiency anaemia?

1. Chronic blood loss
2. Defective iron absorption
3. Defective utilization of iron
4. Inadequate dietary intake
5. Increase erythropoiesis

Which of the following is most useful in distinguishing iron deficiency anaemia from anaemia of chronic illness?

1. Serum iron level
2. Serum ferritin level
3. Serum hepcidin level
4. Total Iron Binding Capacity
5. Transferrin saturation

Clinical Presentations of anaemia

1. Depends on the onset and severity of anaemia
 - Acute and severe: Shortness of breath; palpitation; dizziness or syncope; symptoms associated with cardiac ischaemia
 - Chronic and insidious: Fatigue, decreased exercise tolerance, pale-looking
 - Symptoms related to the causes of anaemia e.g. menorrhagia, passage of tarry stool, bone pain.

Clinical Presentations of anaemia

2. Physical examinations of an anaemic patient

Some examples,

- Uraemic look – anaemia due to kidney failure
- Jaundice – would suggest haemolysis
- Early greying and glossitis – pernicious anaemia
- Shock/tachycardia – acute blood loss
- Isolated splenomegaly - hypersplenism
- Lymphadenopathy and hepatosplenomegaly – haematological malignancy
- Abnormal abdominal mass – watch out for carcinoma

Causes of anaemia

1. Decreased RBC production

Nutritional (Iron, B12, folate)

Bone marrow disorder (e.g. aplastic anaemia, pure red cell aplasia, neoplastic diseases)

Bone marrow suppression (e.g. drugs, chemotherapy, irradiation)

Reduced trophic hormones (e.g. erythropoietin, thyroid hormone)

Anaemia of chronic disease

2. Increased RBC destruction

Intrinsic RBC defects

Enzyme deficiencies (eg, G6PD deficiencies)

Hemoglobin defects (eg, thalassemias, hemoglobinopathies)

Membrane defects (eg, hereditary spherocytosis, elliptocytosis)

Extrinsic RBC defects

Liver disease and hypersplenism

Infections (eg, malaria)

Autoimmune hemolytic anemia (warm- or cold-reacting, drugs)

Microangiopathy (eg, prosthetic valve leak, TTP, DIC)

Transfusion reactions (eg, ABO incompatibility)

3. Blood loss

Gastrointestinal tract, genital tract, soft tissue

Classification of anaemia based on laboratory findings (some examples)

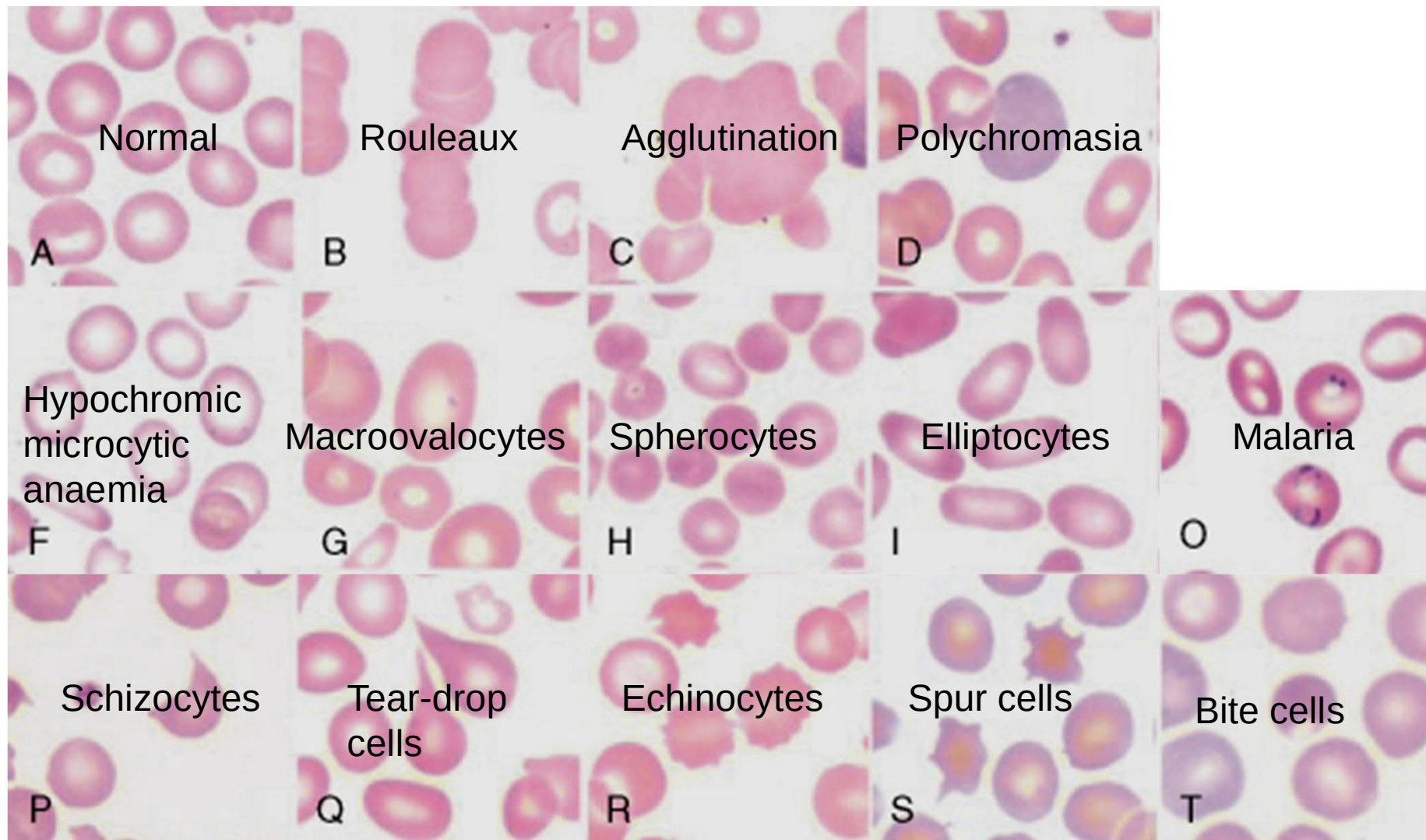
Mean Corpuscular Volume

Low (< 80fL)

Normal (80-99 fL)

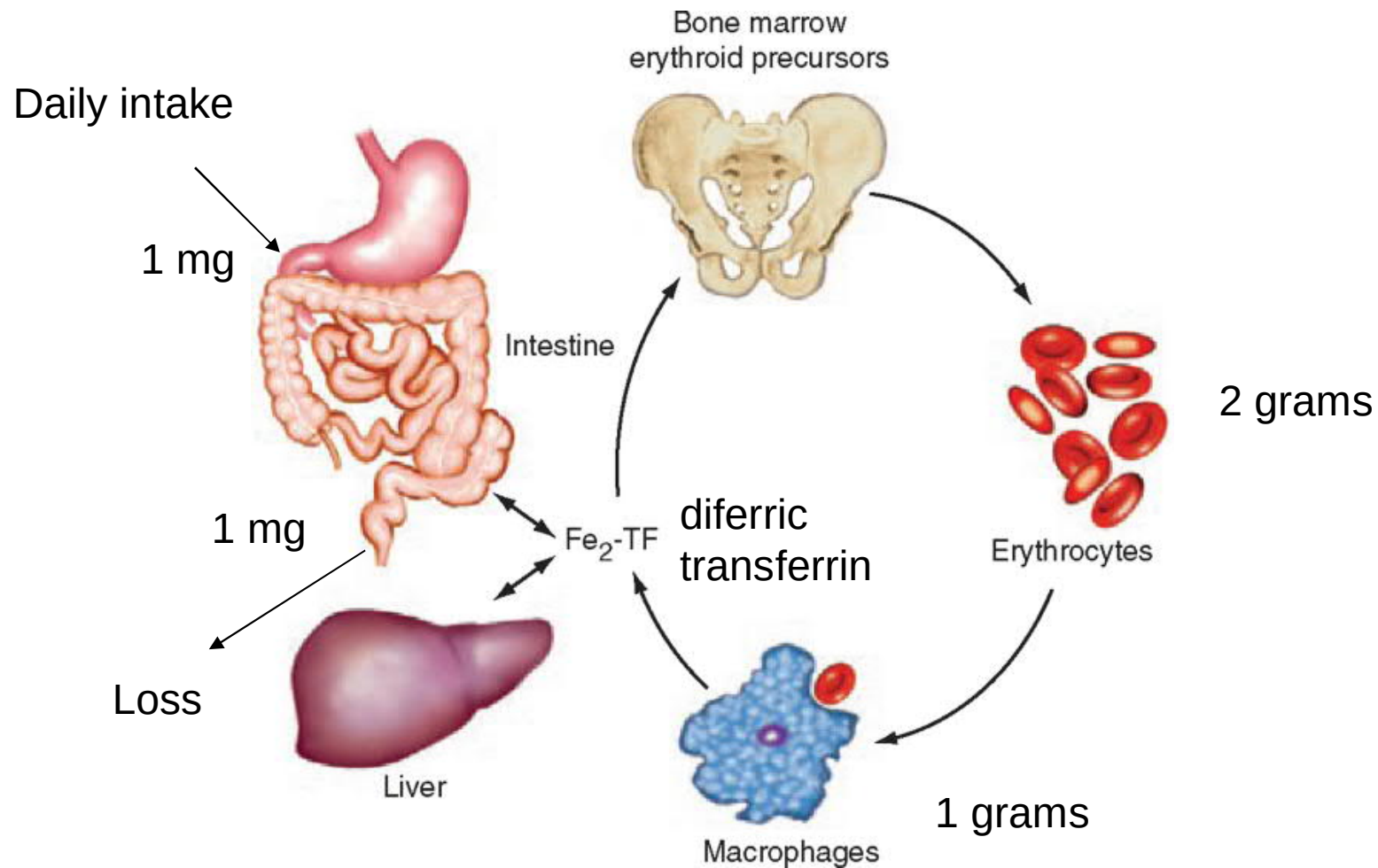
High (>100 fL)

Who's who in the blood film

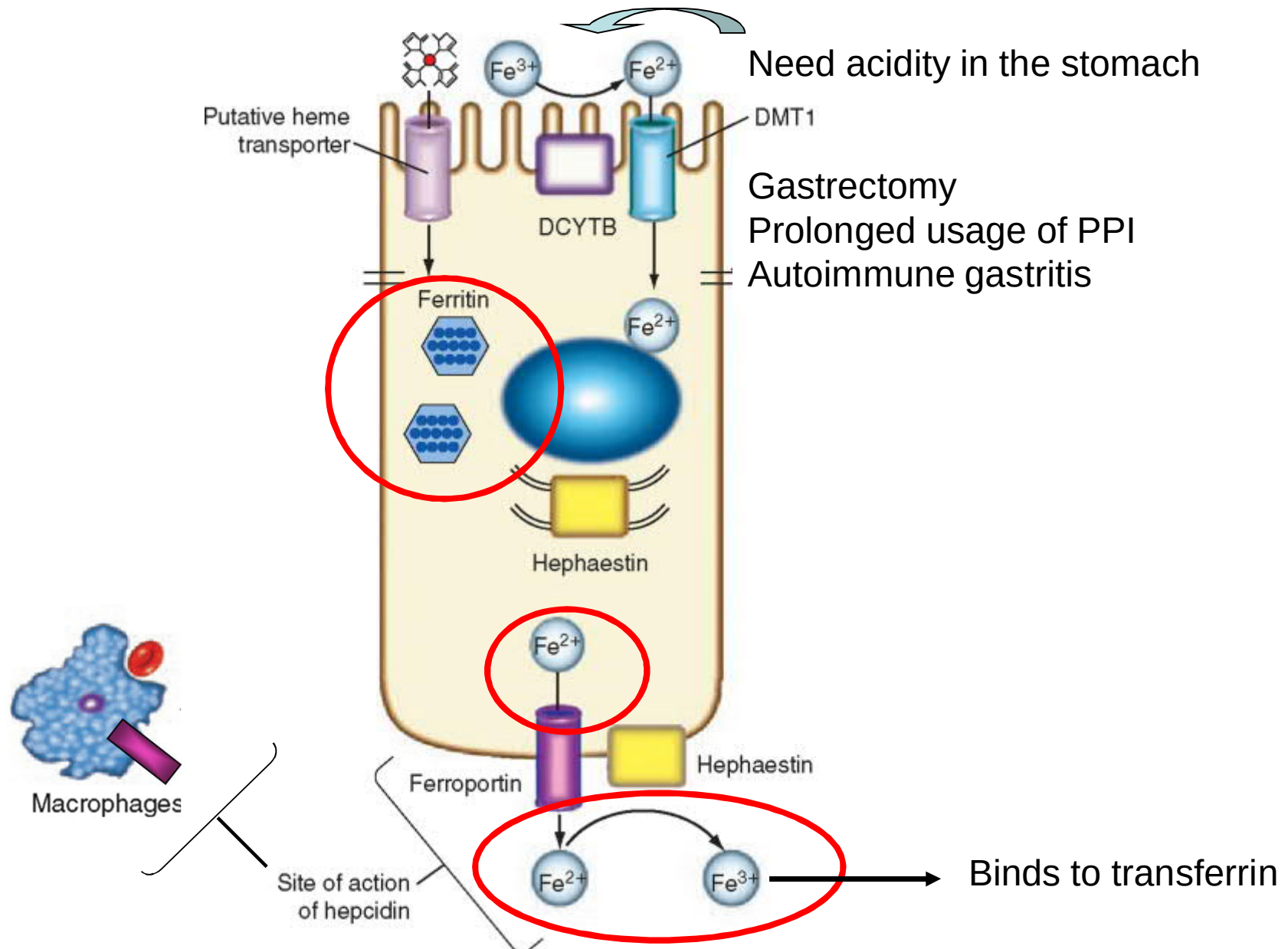


Tear-drop cells, Spur cells, Malaria, Elliptocytes, Normal, Agglutination, Rouleaux, Hypochromic microcytic anaemia, Spherocytes, Schizocytes, Macroovalocytes, Bite cells, Echinocytes, Polychromasia

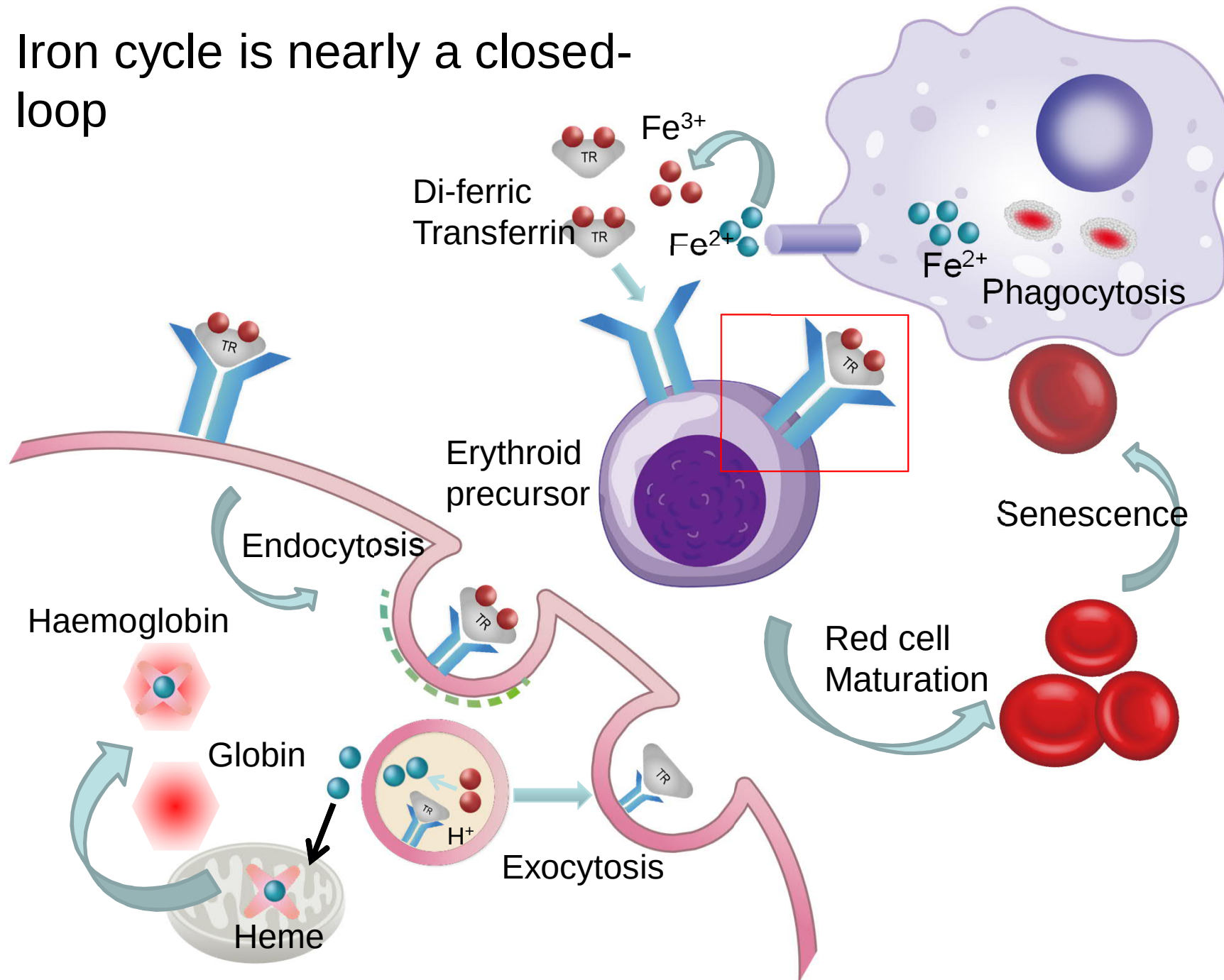
The Iron cycle – almost a closed loop



Absorption of nonheme iron in the duodenal mucosa



Iron cycle is nearly a closed-loop



Examples of iron restricted erythropoiesis

Absolute iron deficiency

- Blood loss via gastrointestinal tract, genital tract.

- Decreased iron absorption due to gastric surgery, malabsorption and atrophic gastritis

- Increased demand as in pregnancy

- Inadequate intake, rare except for instance, anorexia nervosa

Iron-sequestration syndromes

- Anaemia of chronic disease

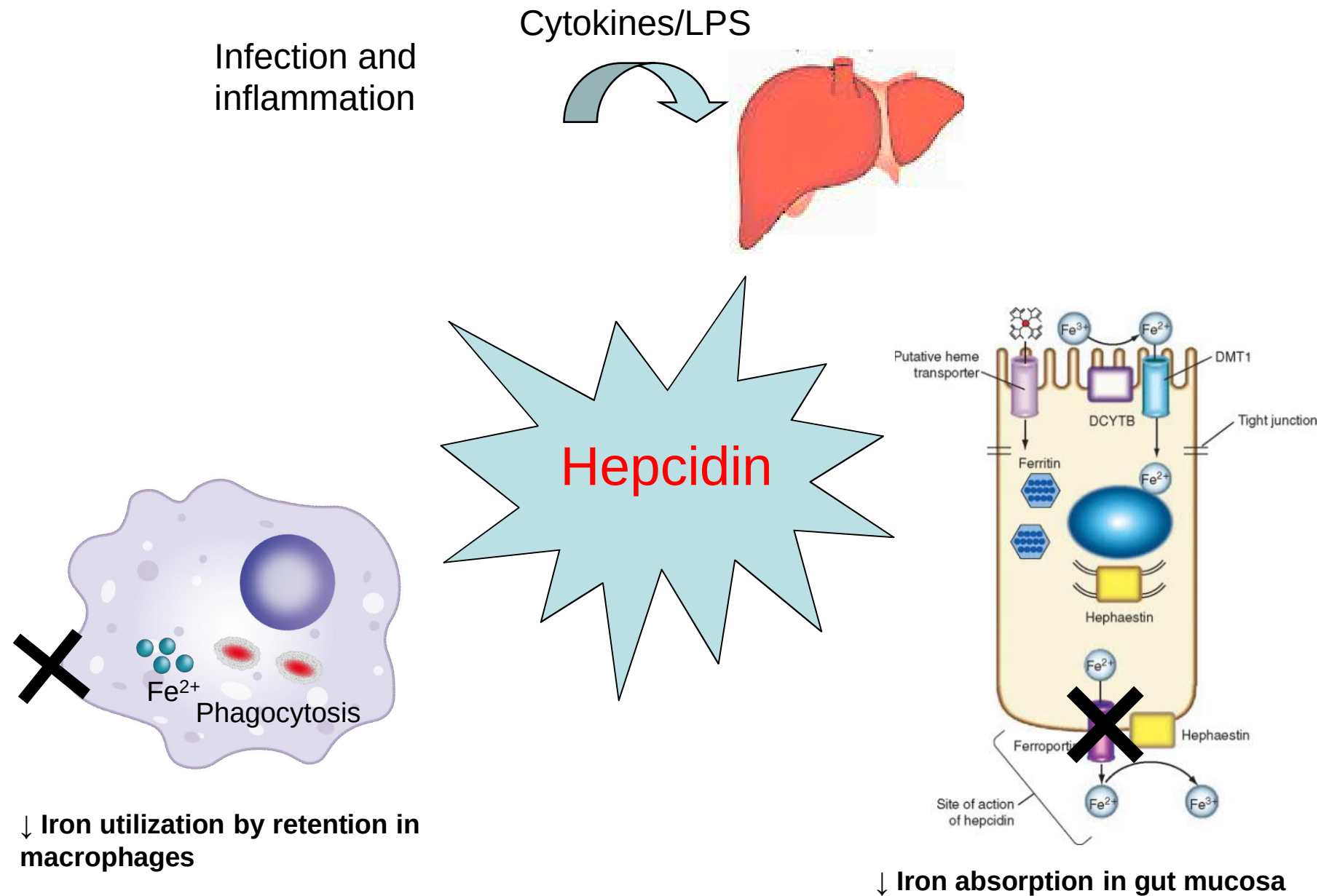
Functional iron deficiency

- Use of erythropoiesis stimulating agents

Anaemia of chronic disease

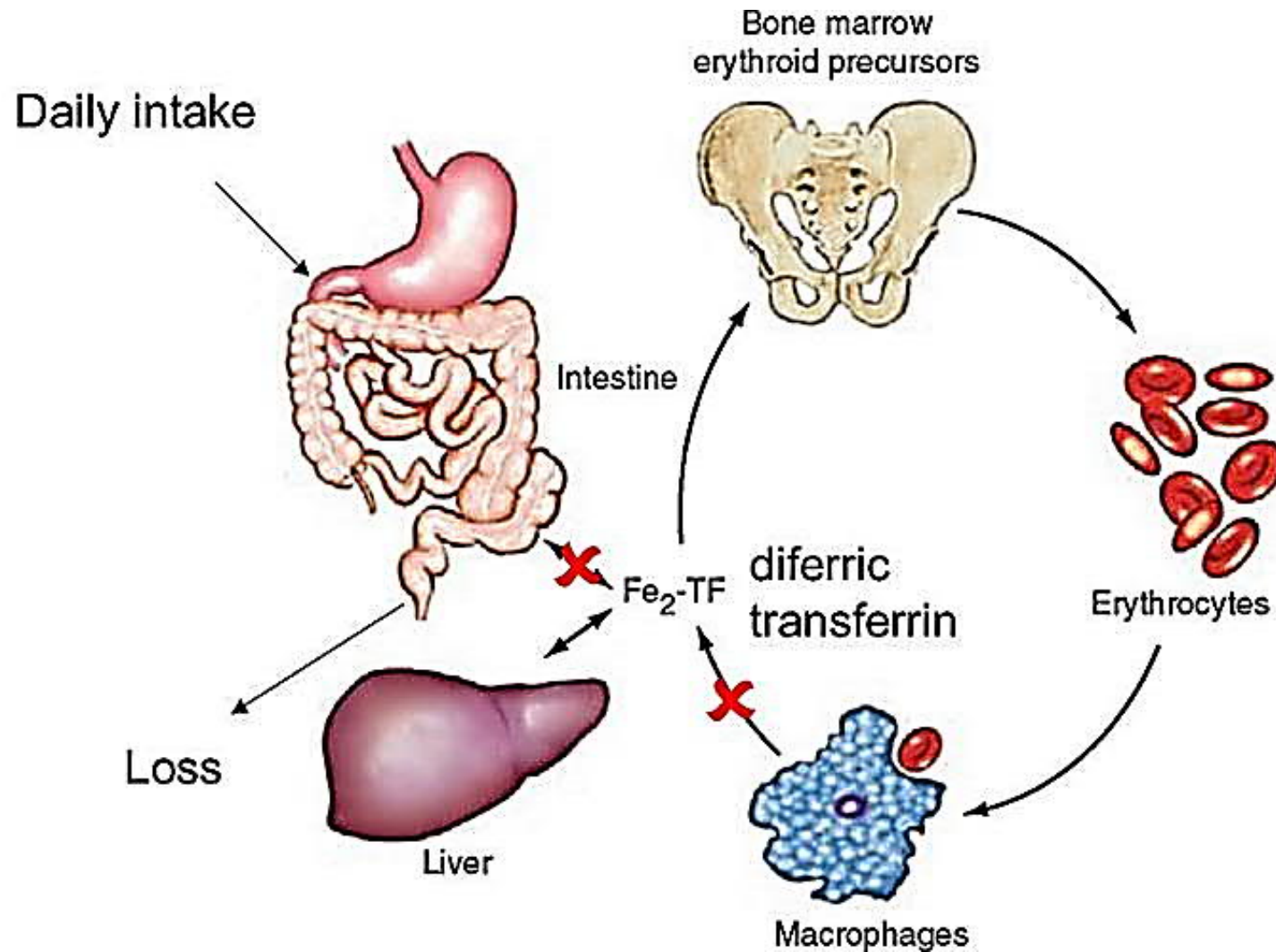
- Infections
- Cancer
- Autoimmune diseases
- Chronic rejection after solid-organ transplantation
- Chronic kidney disease and inflammation

Proposed mechanisms of anaemia of chronic illness



Anaemia of chronic diseases

Hepcidin ✗



Iron deficient anaemia vs anaemia of chronic disease

- Iron deficient anaemia

- Clinical pictures suggesting iron deficiency
- Could be severe
- Hypochromic microcytic
- Reduced serum iron
- Increased transferrin
- Reduced transferrin saturation
- Reduced ferritin
- Elevated soluble transferrin receptor
- Normal serum hepcidin

- Anaemia of chronic disease

- Clinical pictures suggesting chronic disease but not iron deficiency
- Generally modest anaemia
- Normochromic normocytic
- Reduced serum iron
- Decreased/no change in transferrin
- Reduced transferrin saturation
- Normal or increased ferritin
- Normal soluble transferrin receptor
- Elevated serum hepcidin

Absorption of vitamin B12

Sources of Vitamin B12:

1. Foods of animal origin, including fish, meat, poultry, eggs, and dairy products;
2. Fortified cereals and nutritional yeasts
3. Fermented soybean
4. Vitamin B12 supplement

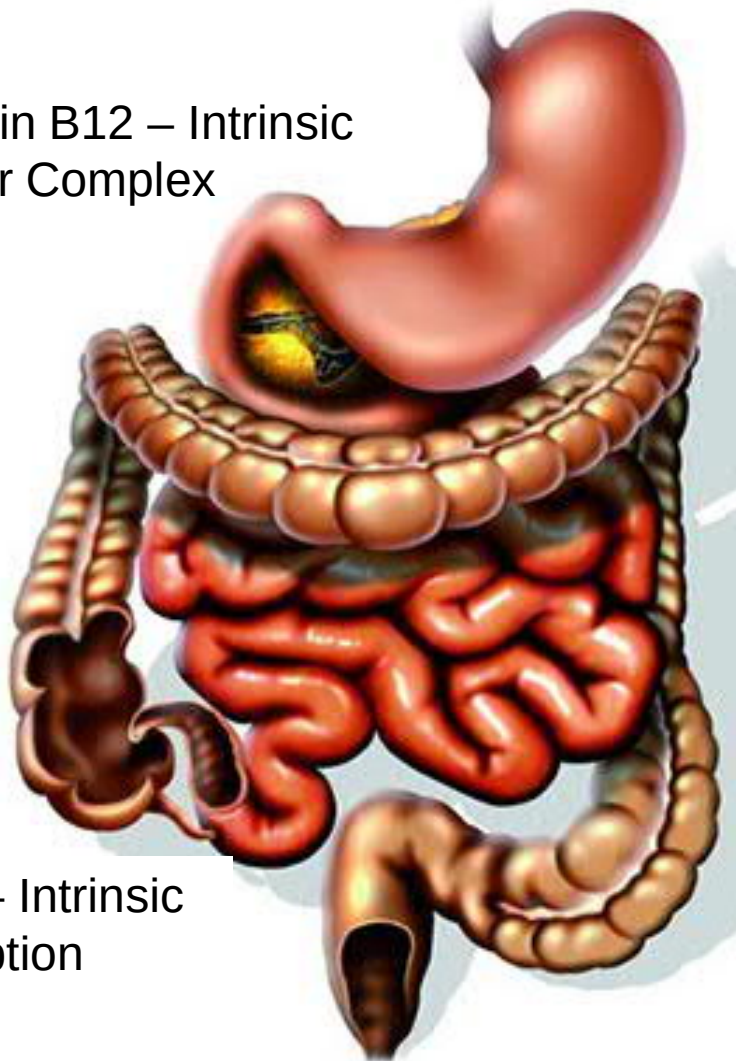
Absorption of vitamin B12

Vitamin B12 in food

Pathophysiology

Vitamin B12 – Intrinsic
Factor Complex

Deficient / defective IF or
release in stomach



Vitamin B12 – Intrinsic
Factor Absorption

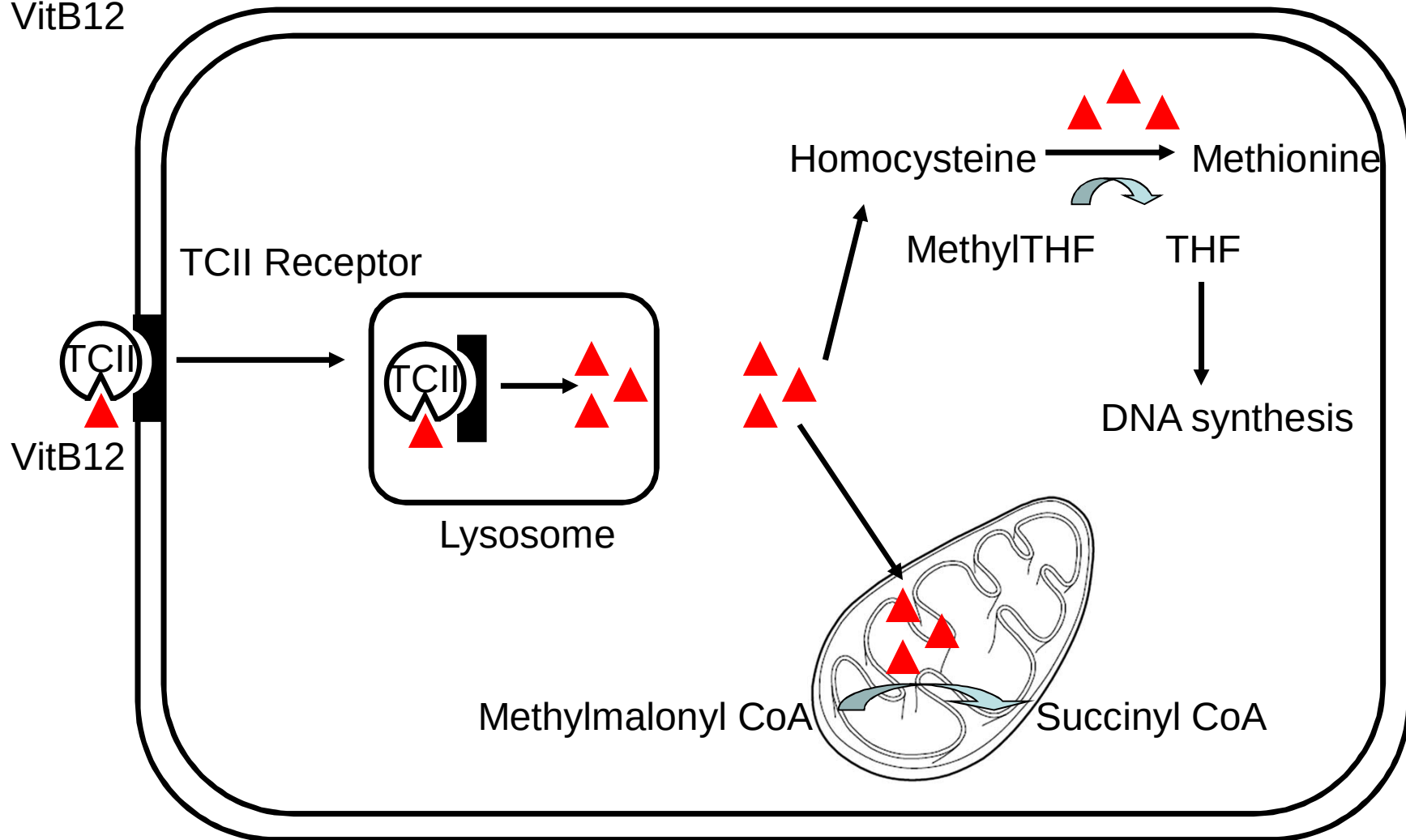
Defective absorption of
Vitamin B12-IF complex at
terminal ileum

Cellular functions of vitamin B12

Haptocorrin



VitB12



Clinical presentation of pernicious anaemia

- Incidental finding of macrocytic anaemia
- Anaemia
- Early greying of hairs
- Mild jaundice
- Glossitis: Beefy-red tongue
- Angular stomatitis
- Neuropathy (subacute combined degeneration of the cord)

Laboratory abnormalities in patients with pernicious anaemia

Pancytopenia

Macrocytic anaemia

Intramedullary hemolysis: Elevated unconjugated bilirubin and LDH

Blood film showed hypersegmented neutrophils and macroovalocytes

Laboratory investigations for patients with suspected pernicious anaemia

Serum Vitamin B12 level

Serum holotranscobalamin

Serum and red cell folate level

Anti-parietal cell antibody – Sensitive but less specific, also detectable in small proportion of normal individuals

Anti-intrinsic factor antibody – Specific but less sensitive, some patients with PA may be negative

Upper endoscopy – Atrophic gastritis and carcinoma of stomach

Schilling test – historical interests, obsolete in HK

Bone Marrow Examination – Not routinely needed, except when laboratory findings incompatible with PA

Case 1

A 70-year-old man presented with malaise and low back pain in recent six months. He was pale.

Laboratory Reports:

Hb: 5.7 g/dL, MCV 88 fL,

Total white cell count: $5.2 \times 10^9/\text{L}$,

Platelet count: $92 \times 10^9/\text{L}$.

Blood smear showed Rouleaux formation.

Case 1

What is the BLOOD test of diagnostic value?

- A. Serum albumin
- B. Serum Fe level
- C. Serum Vit B12 level
- D. Serum globulin
- E. Serum hepcidin

Answer: D

Case 1

The serum globulin level was grossly elevated. What is/are the next diagnostic investigation(s)?

- A. Serum anti-parietal cell antibody
- B. Serum protein electrophoresis
- C. Serum calcium level
- D. Serum lactate dehydrogenase
- E. Serum hepcidin level

Answer: B

Case 2

A 60-year-old woman was admitted to the hospital for dizziness and unsteady gait in recent three months.

Laboratory Reports:

Hb: 5.7 g/dL, MCV 128 fL,

Total white cell count: $2.0 \times 10^9/\text{L}$,

Platelet count: $59 \times 10^9/\text{L}$

Reticulocytes: 3%

Total Bilirubin: 100 $\mu\text{mol/L}$

Lactate Dehydrogenase: 1890 i.u./mL

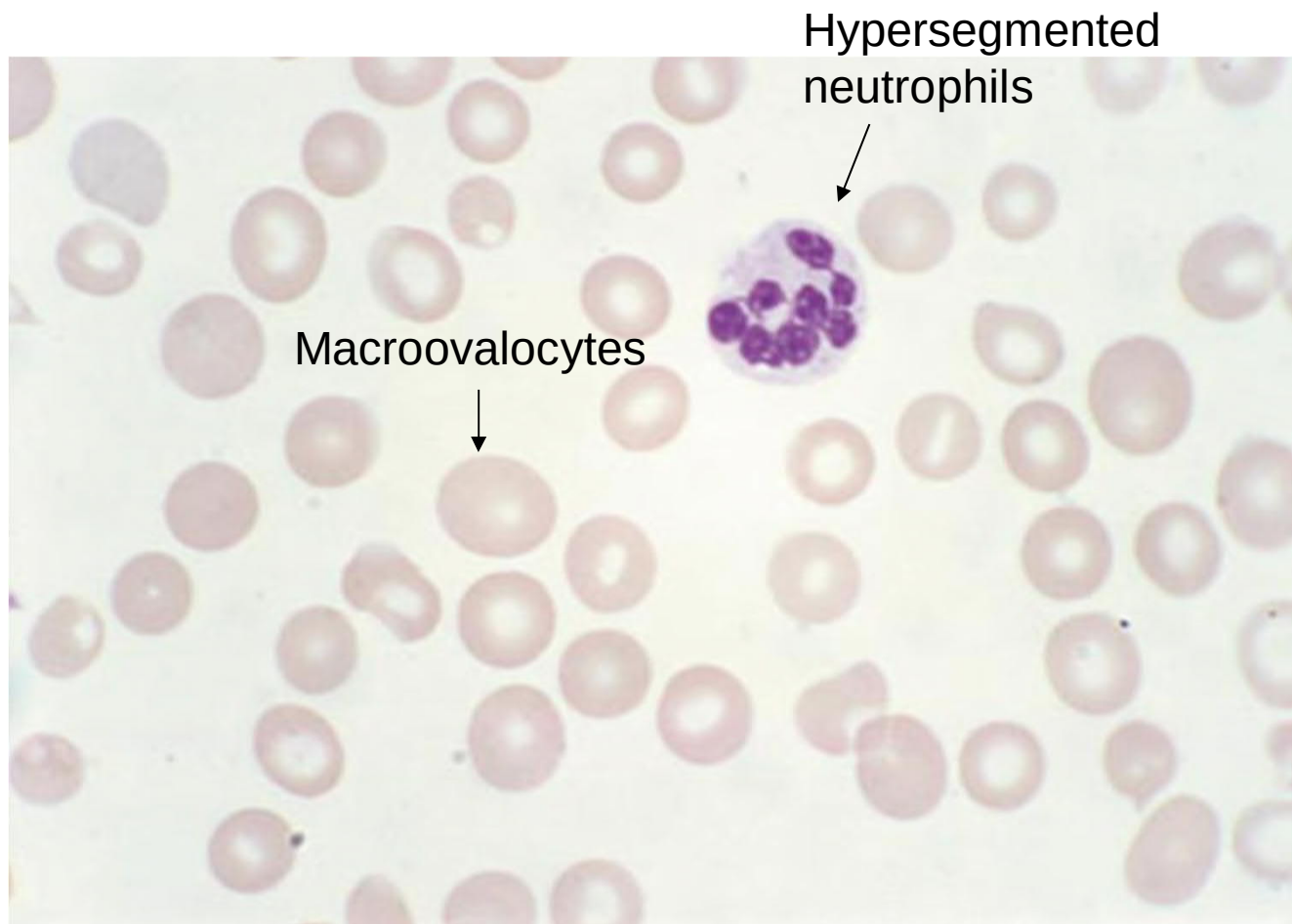
Case 2

What is the next diagnostic investigation?

- A. Peripheral blood smear
- B. Serum Fe level
- C. Thyroid function
- D. Conjugated bilirubin level
- E. Direct Globulin Test

Answer: A

Case 2



Case 2

What is/are the next diagnostic investigation(s)?

- A. Serum B12 level
- B. Serum Intrinsic Factor antibody
- C. Serum and red cell folate level
- D. Serum Parietal Cell antibody
- E. All of the above

Answer: E

Case 2

The diagnosis is pernicious anaemia. What is the next diagnostic investigation?

- A. Bone marrow aspiration
- B. Schilling's Test
- C. Upper endoscopy
- D. Lower endoscopy
- E. All of the above

Answer: C

Case 3

A 60-year-old woman with a metallic mitral valve 20 years ago for underlying chronic rheumatic valve disease was admitted to the hospital for dizziness in recent three months. She was on warfarin at admission.

Laboratory Reports:

Hb: 6.7 g/dL, MCV 99 fL,

Total white cell count: $6.0 \times 10^9/L$,

Platelet count: $259 \times 10^9/L$

Total Bilirubin: 60 $\mu\text{mol/L}$

Lactate Dehydrogenase: 690 i.u./mL

International Normalized Ratio (INR): 2.4

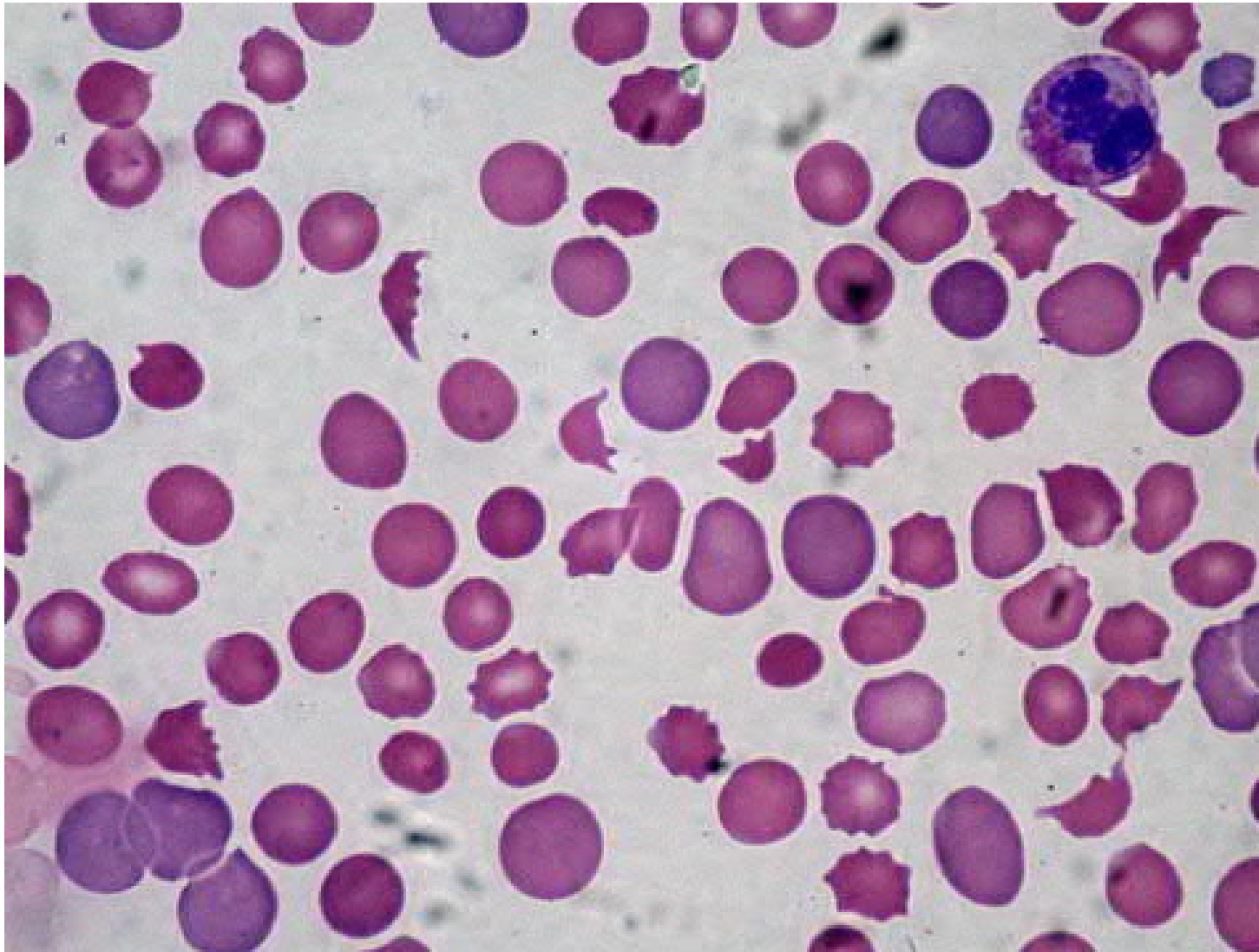
Case 3

What is the next diagnostic investigation?

- A. Peripheral blood smear
- B. Serum Fe level
- C. Serum hepcidin level
- D. Upper endoscopy
- E. Serum ferritin

Answer: A

Case 3



Case 3

What is/are the next diagnostic investigation(s)?

- A. Bone marrow examination
- B. Computer tomography
- C. Upper endoscopy
- D. Lower endoscopy
- E. Echocardiogram

Answer: E

Case 4

A 53-year-old man with history of poorly controlled diabetes mellitus and hypertension presented with malaise and ankle edema in recent six months. He was pale.

Laboratory Reports:

Hb: 6.7 g/dL, MCV 88 fL,

Total white cell count: $6.2 \times 10^9/\text{L}$,

Platelet count: $192 \times 10^9/\text{L}$.

Blood smear showed normochromic normocytic anaemia.

Case 4

What is the BLOOD test of choice?

- A. Serum albumin
- B. Serum Fe level
- C. Serum Vit B12 level
- D. Serum creatinine level
- E. Serum hepcidin

Answer: D

Case 4

His creatinine level was 890 $\mu\text{mol/L}$. What is the likely cause of his anaemia?

- A. Reduced erythropoietin
- B. Elevated serum hepcidin level
- C. Iron deficiency due to occult bleeding
- D. Underlying haematological malignancy
- E. Anaemia due to anti-hypertensive medications

Answer: A

Case 4

He was given weekly replacement of erythropoietin via subcutaneous injection.

Laboratory Reports after three months:

Hb: 7.7 g/dL, MCV 68 fL,

Total white cell count: $5.2 \times 10^9/\text{L}$,

Platelet count: $190 \times 10^9/\text{L}$.

Blood smear showed hypochromic microcytic anaemia.

Case 4

His creatinine level became 990 $\mu\text{mol/L}$. What is/are the likely cause(s) of his persistent anaemia?

- A. Inadequate erythropoietin replacement
- B. Iron deficiency due to occult bleeding
- C. Development of anti-erythropoietin antibody
- D. Functional iron deficiency
- E. Further deterioration of renal function

Answer: B, D

Case 5

A 78 year-old man with history of right knee pain due to osteoarthritis complained of malaise, shortness of breathe and dizziness for three months.

Laboratory Reports

- Haemoglobin: 8.5 g/dL;
- Mean corpuscular volume: 70 fl;
- Total white cell count: $14 \times 10^9/\text{L}$;
- Platelet count: $719 \times 10^9/\text{L}$.
- Blood film showed hypochromic microcytic anaemia with a few pencil cells

Case 5

What is/are the next diagnostic investigation(s)?

- A. Serum B12 level
- B. Serum iron level
- C. Serum ferritin level
- D. Serum hepcidin
- E. Serum folate level

Answer: B

Case 5

He was found to have low levels of serum iron and transferrin saturation and raised TIBC.

What is/are the next investigation(s)?

- A. Upper endoscopy to look for bleeding
- B. Serum hepcidin to rule out anaemia of chronic disease
- C. Bone marrow examination to look for underlying myeloproliferative neoplasm
- D. Stool for occult blood to look for occult GI bleeding
- E. Both A and C

Answer: A

Case 5

Upper endoscopy showed bleeding gastritis

What is the cause of this problem?

- A. *Helicobacter pylori* infection
- B. Underlying adenocarcinoma of stomach
- C. Intake of non-steroid anti-inflammatory drug for underlying osteoarthritis
- D. Bleeding tendency related to thrombocytosis
- E. All of the above

Answer: C

Case 6

A 19 year-old male immigrant from China complained of short stature and worsening malaise.

Laboratory Reports

- Haemoglobin: 7.5 g/dL;
- Mean corpuscular volume: 103 fl;
- Total white cell count: $2.0 \times 10^9/\text{L}$;
- Platelet count: $109 \times 10^9/\text{L}$.
- Blood film showed macrocytic anaemia with a few dysplastic neutrophils

Case 6



Case 6

What is/are the relevant history in this patient?

- A. Where was he born in China?
- B. Has his mother been exposed to toxin?
- C. Was his father similarly affected?
- D. Were his siblings similarly affected?
- E. Any history of child abuse?

Answer: D

Case 6

What is the diagnostic test for this patient?

- A. Bone marrow examination
- B. Examination of his siblings
- C. Demonstration of chromosome breaks
- D. Skin Biopsy
- E. None of the above

Answer: C

Case 6

What is the most worrisome condition of this disease?

- A. Leukemia transformation
- B. Malignant transformation of skin
- C. Cardiac arrhythmia
- D. Neurological degeneration
- E. All of the above

Answer: A